SOGx2 EVB StarterKit-QSG

Part Number	902-SOG42-Vx0RH / 902-SOG52-Vx0RH / 902-SOG72-Vx0RH
Platform Form	MediaTek MT8371LV / MT8371 / MT8391
Product Name	SOGx2 EVB StarterKit
Description	SOG42/52/72 Evaluation Kit based on MediaTek Genio 420/520/720
Revision	V1.1
Issue Date	20260713

Approved by	Reviewed by	Issued by
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Revision History

Version	Date	Description
0.1	2026/05/06	draft
1.0	2026/05/15	Release
1.1	2026/07/13	Camera module info revised. Added SOG42 info. Overview, additional items need for QSG Determining the OSM Storage Type Content refinement

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SOGx2 EVB StarterKit Overview

The SOGx2 EVB StarterKit is engineered for embedded applications that require reliable, low-power implementation of edge AI and GenAI applications. Built around the advanced 6nm MediaTek Genio SoC family, the kit is available in three configurations: **SOG42 (powered by the MT8371LV / Genio 420), SOG52 (powered by the MT8371 / Genio 520), and SOG72 (powered by the MT8391 / Genio 720)**. The platform integrates an 8th generation MediaTek NPU that delivers up to 10 TOPS of acceleration for handling complex AI tasks like object detection, image classification, and speech recognition, as well as on-device support for GenAI applications and leading LLMs (Large Language Models). This enables real-time, low-latency AI inference on the edge, reducing reliance on cloud-based processing.

The SOGx2 platform also features an octa-core processor architecture with 2x Arm Cortex-A78 cores and 6x Arm Cortex-A55, paired with an Arm Mali-G57 MC2 GPU. It supports up to dual displays, or a single display up to 4K60, making it a strong fit for smart retail, smart home, transportation/smart cities, healthcare, entertainment, education, industrial, and enterprise applications.

The SOGx2 StarterKit relies on an OSM (Open Standard Module) reference design and is pin-to-pin and software compatible across the MT8371LV, MT8371, and MT8391 models, allowing for seamless migration depending on your computing and budget requirements. With comprehensive support for Android, IoT Yocto, and Ubuntu, it provides a stable foundation for long-lifecycle IoT deployments.

SOGx2 EVB StarterKit Requirements

This EVB StarterKit supports a wide range of hardware configurations and Android software environments. For simplicity, this Quick Start Guide will demonstrate a specific setup using the following conditions:

- EVB: SOGx2 EVB
- SoM: OSM42 / 52 / 72
- Boot Mode: eMMC Boot
- Display Mode : DP
- Operating System: Android 16
- Host PC: Windows 10 / Ubuntu 22.04

Additional Items Needed for Quick Start Guide Demo

To support the demo, the following additional items are needed.

- USB Type-C to USB Type-A adapter
- USB keyboard
- USB mouse
- DisplayPort (DP) monitor
- DisplayPort (DP) cable

Setup Host PC Environment

Install a terminal emulator on your Linux or Windows PC. This guide uses **Tera Term 5.4.0** on Windows 10 as an example. For detailed settings, please refer to the "Connect to the Serial Console" section.

SOGx2 EVB StarterKit Hardware Overview

SOGx2 EVB mainboard

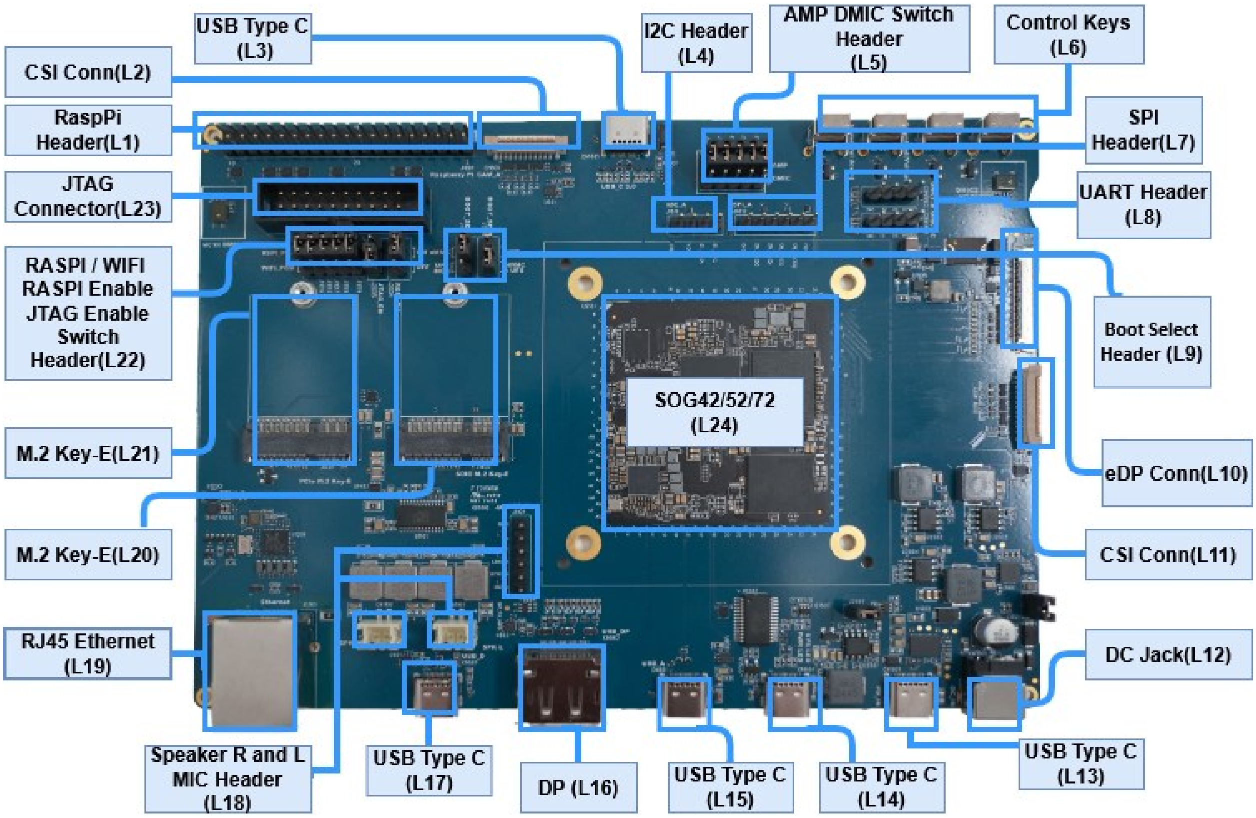


Figure 1 Top view

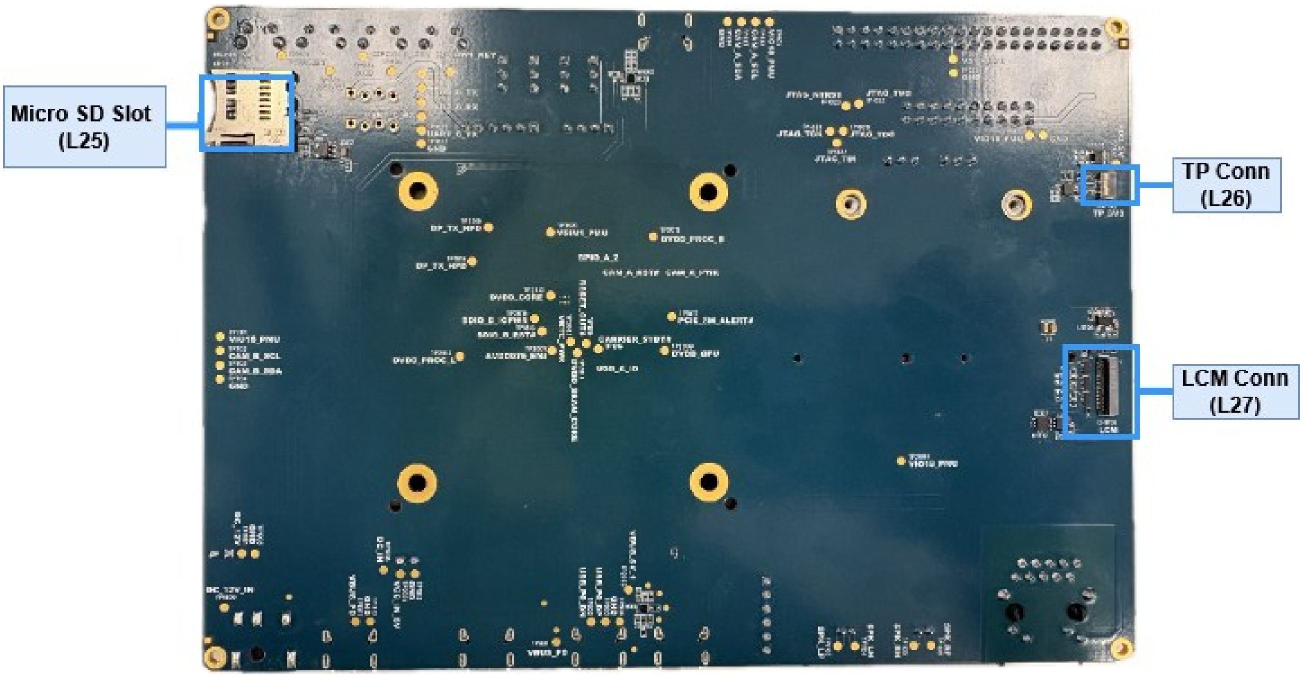
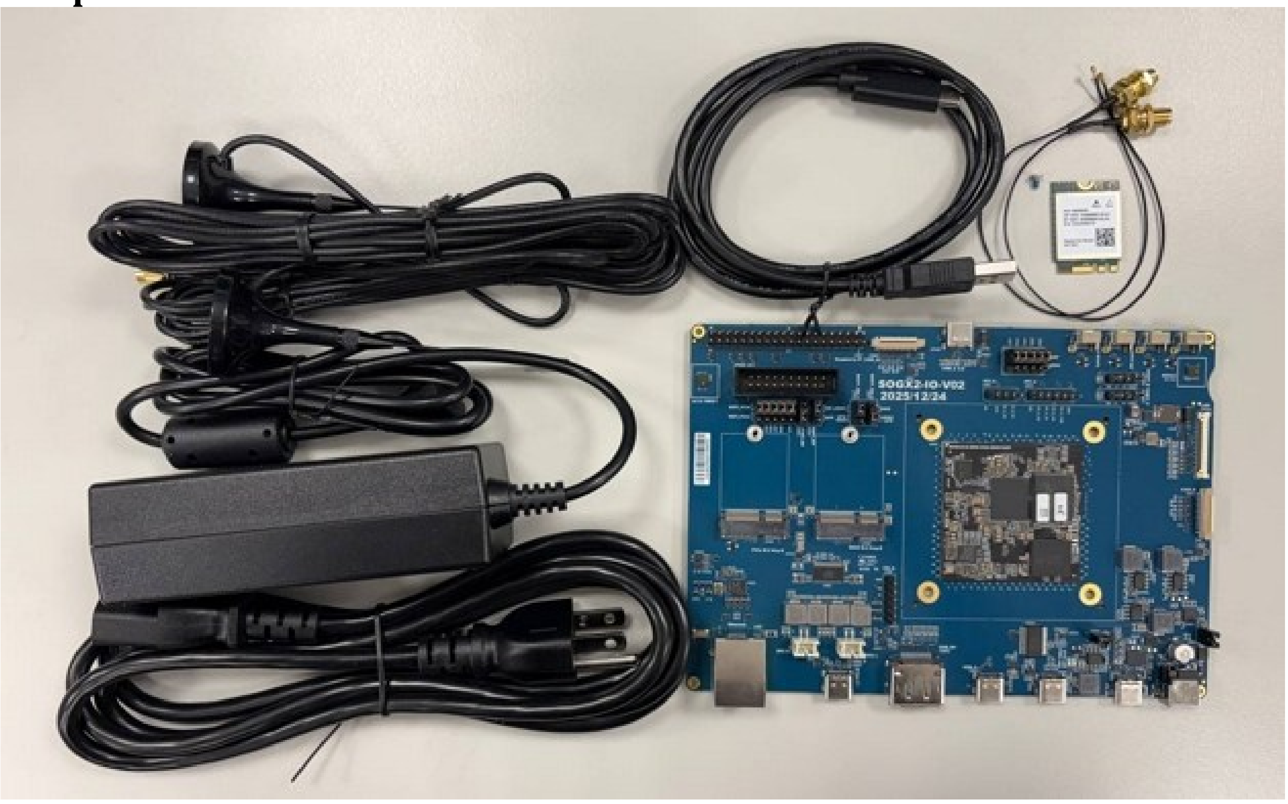


Figure 2 Bottom

SOGx2 EVB Packing List

No.	Part Number	Item Name	Description	Quantity
1	902-SOG42-Vx0RH / 902-SOG52-Vx0RH / 902-SOG72-Vx0RH	SOG42/52/72 EVK Main Board	ASSY BOARD SOG42/SOG52/SOG72 EVK Main Board	1
2	314-SSBE8-US0A8	12V DC JACK Power Adapter	AC ADAPTER 12V/5A + AC POWER CORD C13 US (T152)	1
3	306-SOG52-000XX	IPEX wire	COAXIAL CABLE SMA TO IPEX4 OD1.32*L200mm	2
4	087-SSBE6-W001G/ 087-UT180-A00EP	Antenna Cable set	SMA ANTENNA cable set	2
5	309-SOG52-000ER	Debug cable	USB A to C CABLE L=1M (**Debug cable not for charging, only for data transfer**)	1
6	007-SSBE8-W00A6	AW-XB468NF	WiFi 6 module-PCIe AW-XB468NF (7921)	1
7	203-M2044-021MD	Screw	Screw for assemble AW-XB468NF	1
8	413-SOG36-000XX	User Manual	User Manual with data sheet download	1

Peripherals Picture



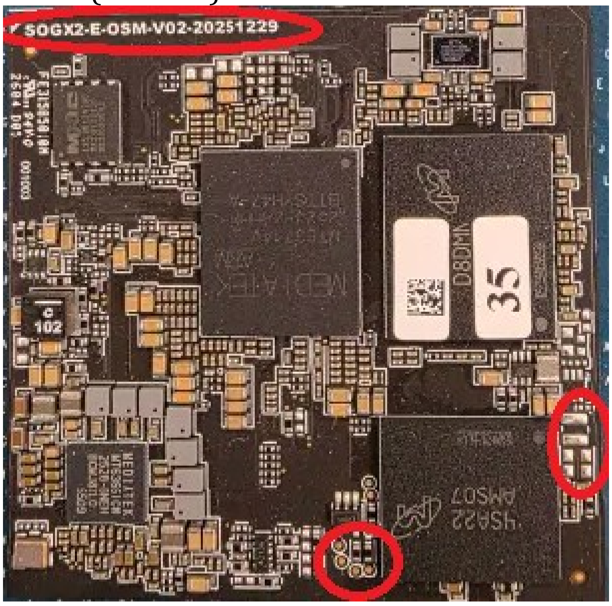
SOGx2 EVB StarterKit Installation and Configuration

Boot Mode Select Jumper Configuration

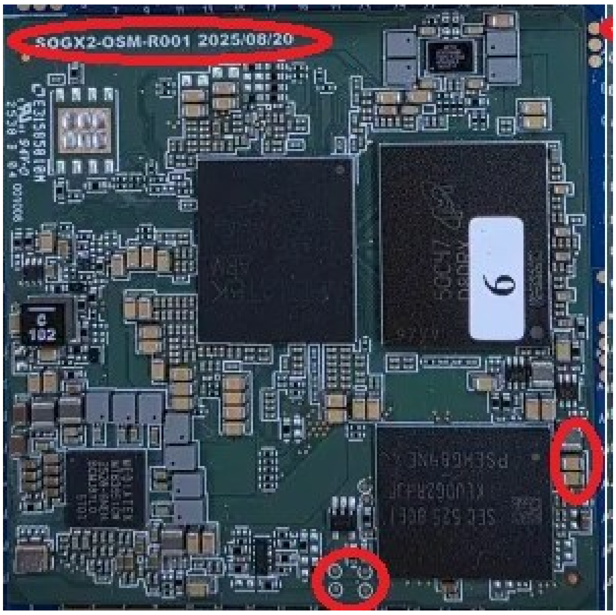
Before proceeding with any boot or flashing operation, always confirm the target ROM (storage) device.

Determining the OSM Storage Type:

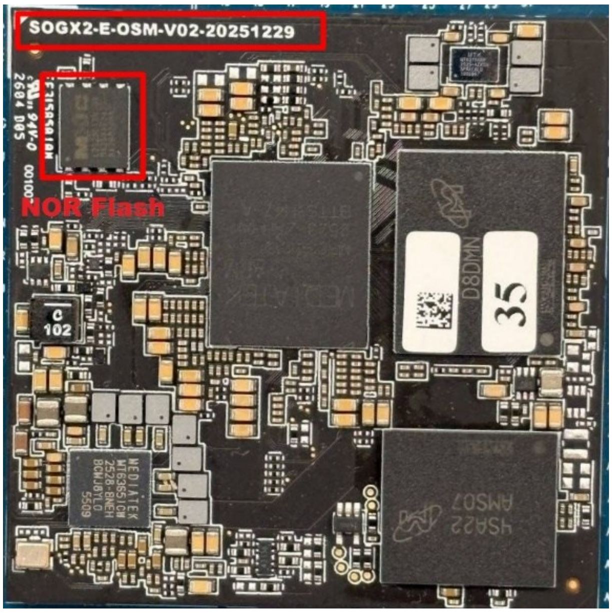
eMMC(**Default**) SOGX2-E-OSM-V02



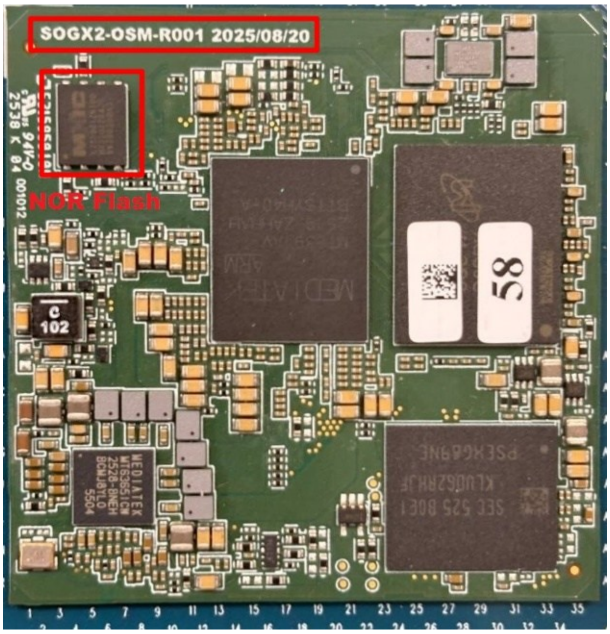
UFS : SOGX2-OSM-R001



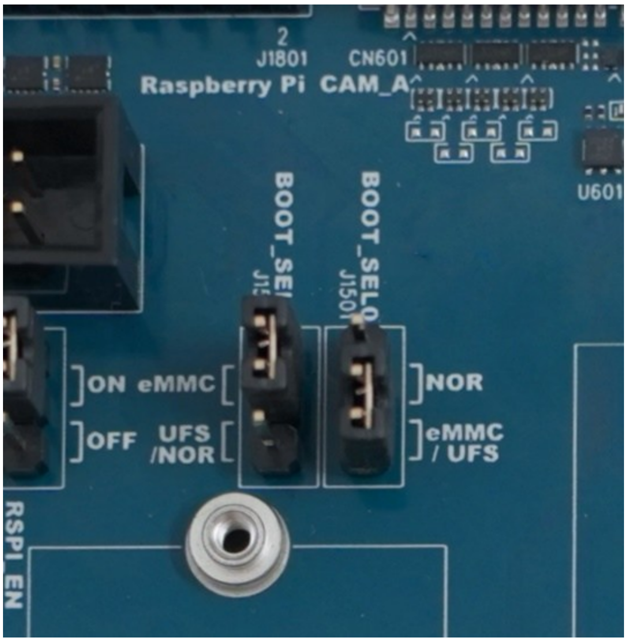
eMMC with NOR Flash : SOGX2-E-OSM-V02



UFS with NOR Flash : SOGX2-OSM-R001



- **eMMC:** The SOG42/52/72 EVK uses eMMC as the primary boot and storage device.
- If using a storage device other than eMMC, set the boot selection jumper (L9) accordingly.
    - **NOR:** Set **BOOT_SEL0 & BOOT_SEL1** jumper to NOR
    - **UFS:** Set **BOOT_SEL0 & BOOT_SEL1** jumper to UFS



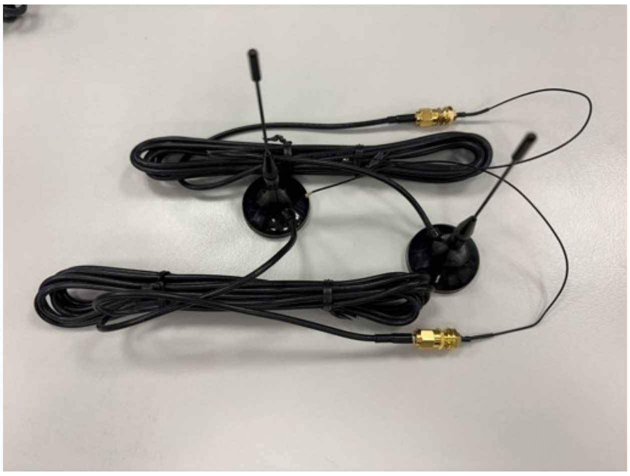
Function	Description		
Location	**L9 (J1502 & J1501)**		
Boot Mode	**J1502 Setting (BOOT_SEL1)**	**J1501 Setting (BOOT_SEL0)**	**Description**
NOR	Lower (Pin 2-3)	Upper (Pin 1-2)	System boots from NOR Flash.
eMMC (Default)	**Upper (Pin 1-2)**	**Lower (Pin 2-3)**	**System boots from eMMC.**
UFS	Lower (Pin 2-3)	Lower (Pin 2-3)	System boots from UFS.

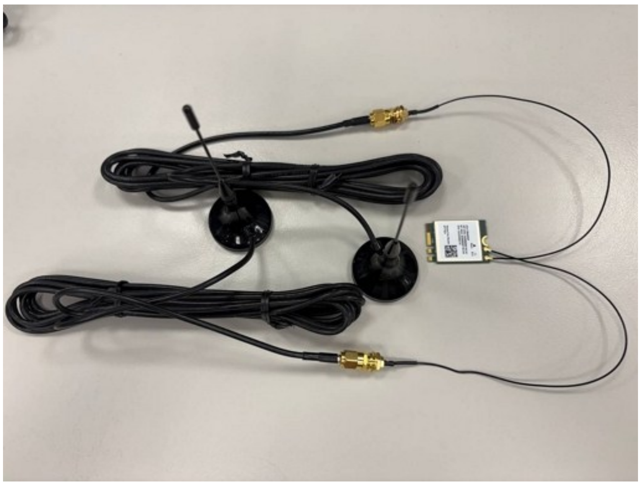
For all other jumper configurations and functions (AMP DMIC Switch & RASPI / WIFI & RASPI Enable & JTAG Enable Switch etc.), please refer to the file: SOGx2-SOM-Carrier Board Hardware User Guide.

Note: NOR boot is supported only for Yocto OS.

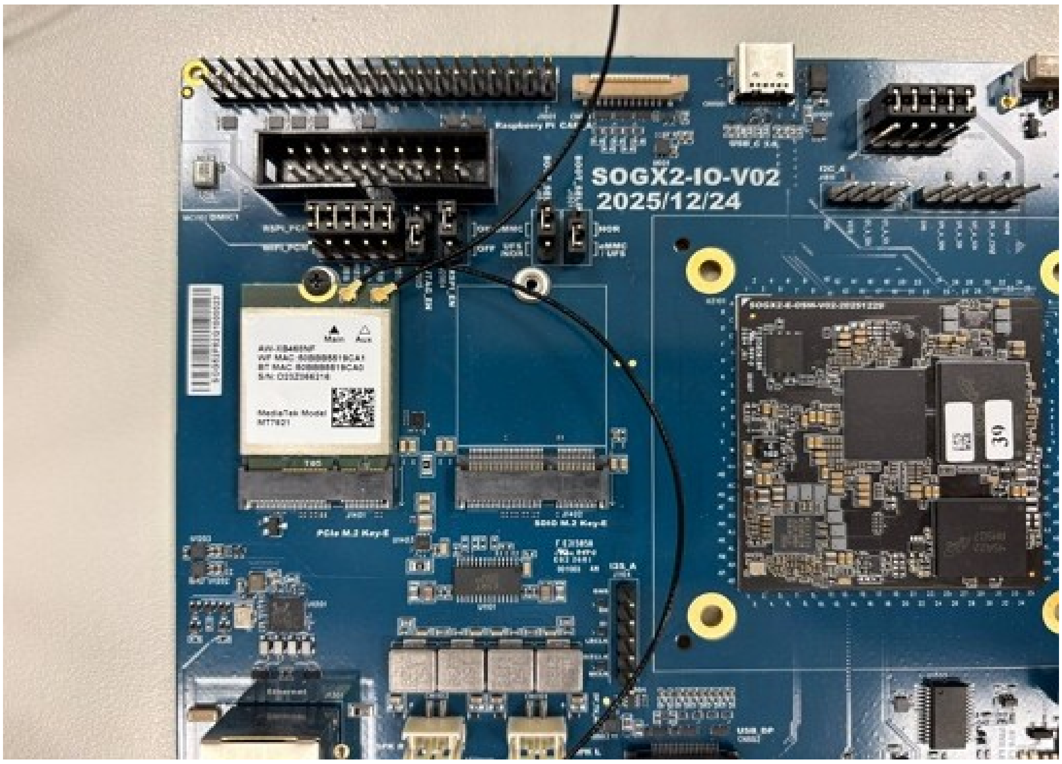
WiFi Module Installation

Insert the Wi-Fi module into the PCIe M.2 Key-E slot(L21), secure it with the screw (Packing list #7: 203-M2044-021MD), and install the Wi-Fi antennas.





Assemble WiFi module with mainboard



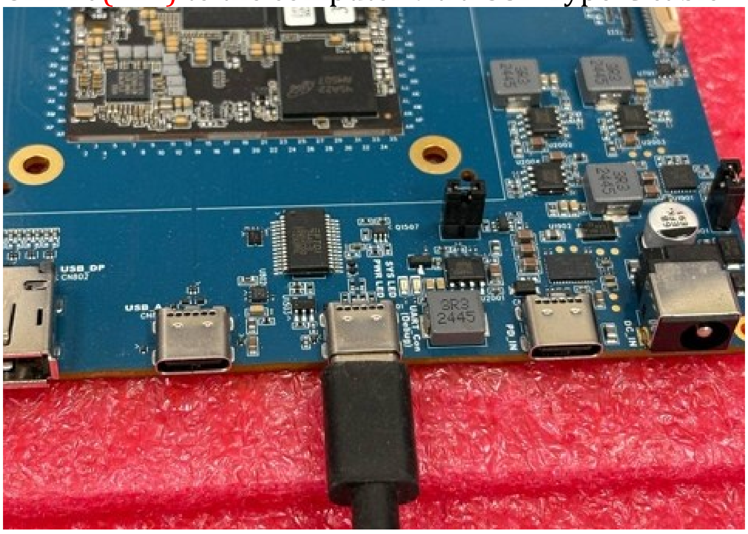
Connect to the Serial Console

Set up the serial port on the host PC to view debug output and interact with the Starter Kit through a command-line interface.

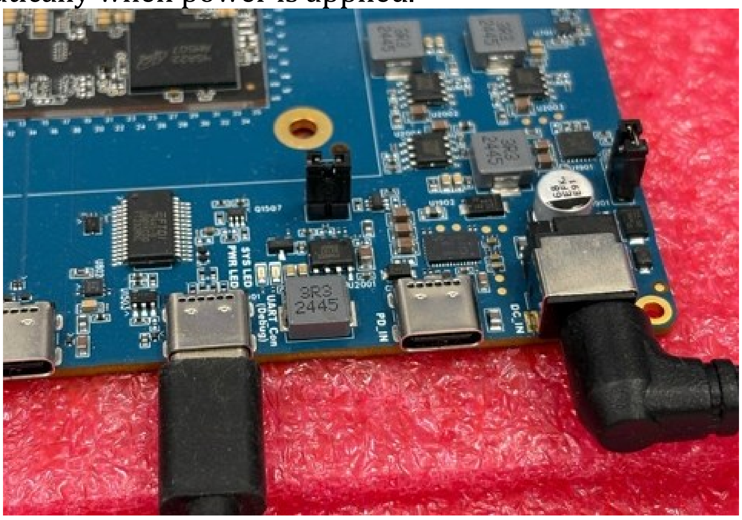
- **Windows:** Tera Term (5.4.0) / Putty
- **Linux:** Minicom / Picocom

Install procedure for viewing EVB StarterKit logs

- **Step 1:** Connect UART0(L14) to the computer via a USB Type-C cable.

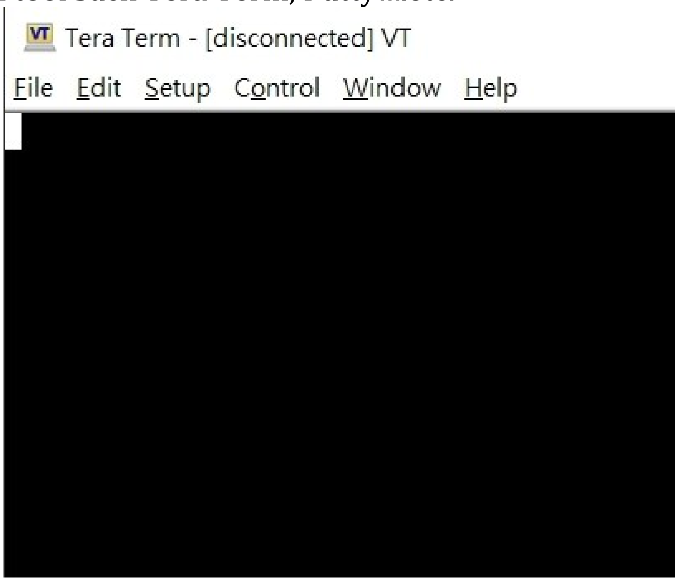


- **Step 2:** Connect the DC jack power adapter, but do not apply power yet. The system will boot automatically when power is applied.

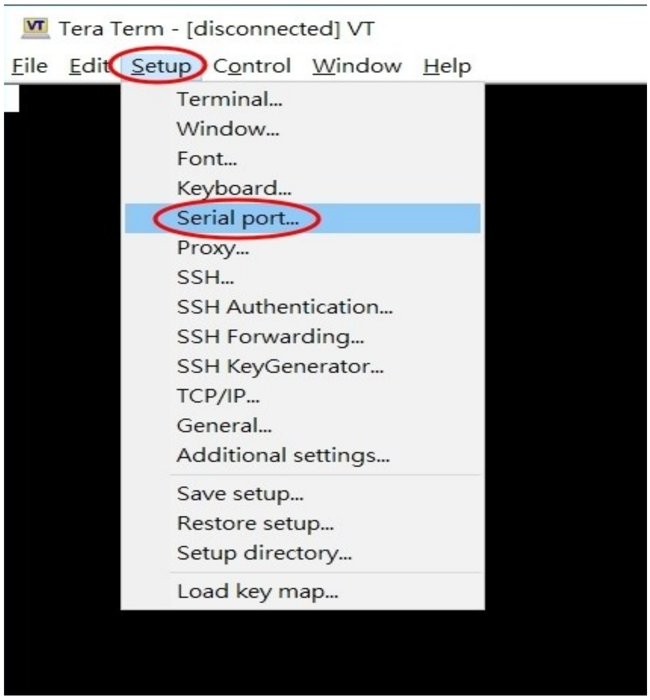


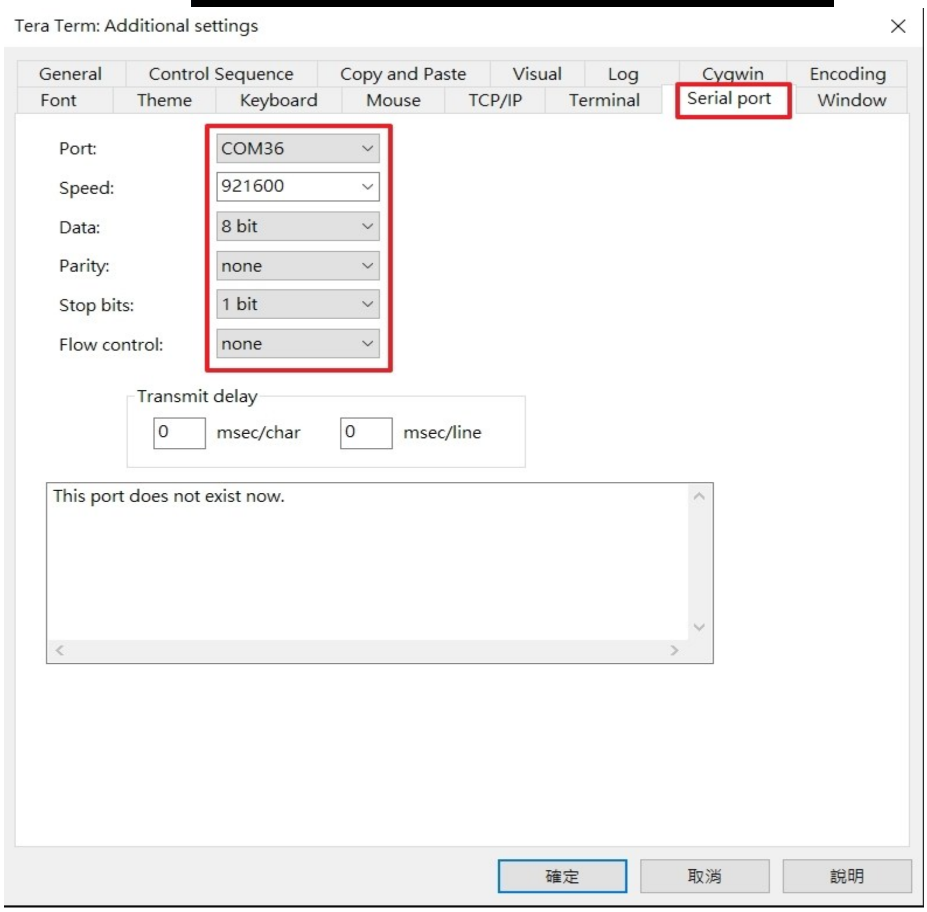
- **Step 3:** Check what Serial port number device that used.

- **Step 4:** Open serial tool such Tera Term, Putty….etc.

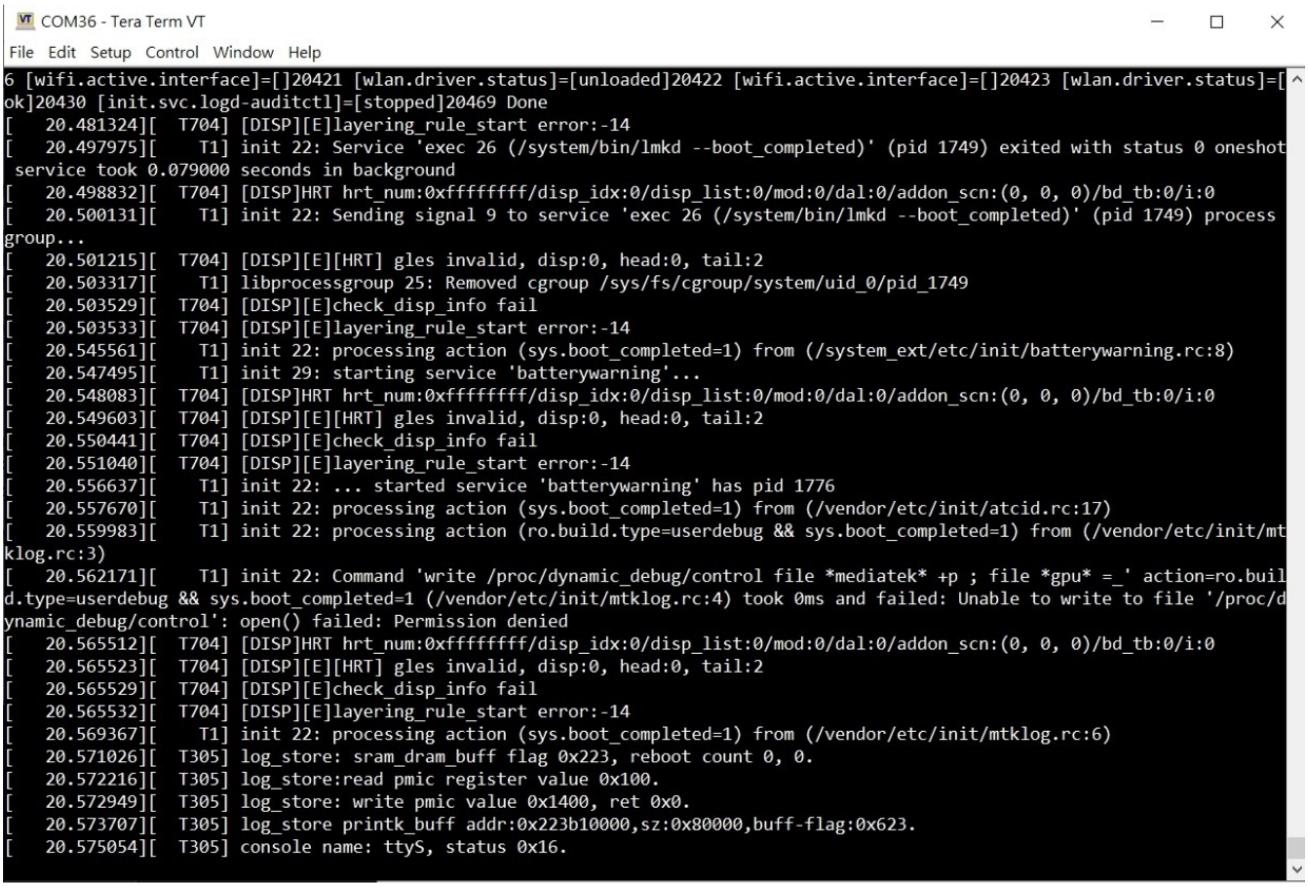


- **Step 5:** To configure the UART port settings, specify the following parameters:
  Serial Port Number: Select the appropriate serial port number
  Baud Rate: Set the baud rate to 921600
  Data Bits/Parity/Stop Bits: Configure as 8 data bits, None parity, and 1 stop bit





Then connect the monitor to the SOGx2 EVB using a DP cable. Once all connections are complete, apply power to the board. The system will boot automatically. You should see the display output on the monitor and the trace logs printed through UART0 on the host PC.



```
COM36 - Tera Term VT
File  Edit  Setup  Control  Window  Help
6 [wifi.active.interface]=[]20421 [wlan.driver.status]=[unloaded]20422 [wifi.active.interface]=[]20423 [wlan.driver.status]=[
ok]20430 [init.svc.logd-auditctl]=[stopped]20469 Done
[   20.481324][  T704] [DISP][E]layering_rule_start error:-14
[   20.497975][    T1] init 22: Service 'exec 26 (/system/bin/lmkd --boot_completed)' (pid 1749) exited with status 0 oneshot
 service took 0.079000 seconds in background
[   20.498832][  T704] [DISP]HRT hrt_num:0xffffffff/disp_idx:0/disp_list:0/mod:0/dal:0/addon_scn:(0, 0, 0)/bd_tb:0/i:0
[   20.500131][    T1] init 22: Sending signal 9 to service 'exec 26 (/system/bin/lmkd --boot_completed)' (pid 1749) process
group...
[   20.501215][  T704] [DISP][E][HRT] gles invalid, disp:0, head:0, tail:2
[   20.503317][    T1] libprocessgroup 25: Removed cgroup /sys/fs/cgroup/system/uid_0/pid_1749
[   20.503529][  T704] [DISP][E]check_disp_info fail
[   20.503533][  T704] [DISP][E]layering_rule_start error:-14
[   20.545561][    T1] init 22: processing action (sys.boot_completed=1) from (/system_ext/etc/init/batterywarning.rc:8)
[   20.547495][    T1] init 29: starting service 'batterywarning'...
[   20.548083][  T704] [DISP]HRT hrt_num:0xffffffff/disp_idx:0/disp_list:0/mod:0/dal:0/addon_scn:(0, 0, 0)/bd_tb:0/i:0
[   20.549603][  T704] [DISP][E][HRT] gles invalid, disp:0, head:0, tail:2
[   20.550441][  T704] [DISP][E]check_disp_info fail
[   20.551040][  T704] [DISP][E]layering_rule_start error:-14
[   20.556637][    T1] init 22: ... started service 'batterywarning' has pid 1776
[   20.557670][    T1] init 22: processing action (sys.boot_completed=1) from (/vendor/etc/init/atcid.rc:17)
[   20.559983][    T1] init 22: processing action (ro.build.type=userdebug && sys.boot_completed=1) from (/vendor/etc/init/mt
klog.rc:3)
[   20.562171][    T1] init 22: Command 'write /proc/dynamic_debug/control file *mediatek* +p ; file *gpu* =_' action=ro.buil
d.type=userdebug && sys.boot_completed=1 (/vendor/etc/init/mtklog.rc:4) took 0ms and failed: Unable to write to file '/proc/d
ynamic_debug/control': open() failed: Permission denied
[   20.565512][  T704] [DISP]HRT hrt_num:0xffffffff/disp_idx:0/disp_list:0/mod:0/dal:0/addon_scn:(0, 0, 0)/bd_tb:0/i:0
[   20.565523][  T704] [DISP][E][HRT] gles invalid, disp:0, head:0, tail:2
[   20.565529][  T704] [DISP][E]check_disp_info fail
[   20.565532][  T704] [DISP][E]layering_rule_start error:-14
[   20.569367][    T1] init 22: processing action (sys.boot_completed=1) from (/vendor/etc/init/mtklog.rc:6)
[   20.571026][  T305] log_store: sram_dram_buff flag 0x223, reboot count 0, 0.
[   20.572216][  T305] log_store:read pmic register value 0x100.
[   20.572949][  T305] log_store: write pmic value 0x1400, ret 0x0.
[   20.573707][  T305] log_store printk_buff addr:0x223b10000,sz:0x80000,buff-flag:0x623.
[   20.575054][  T305] console name: ttyS, status 0x16.
```

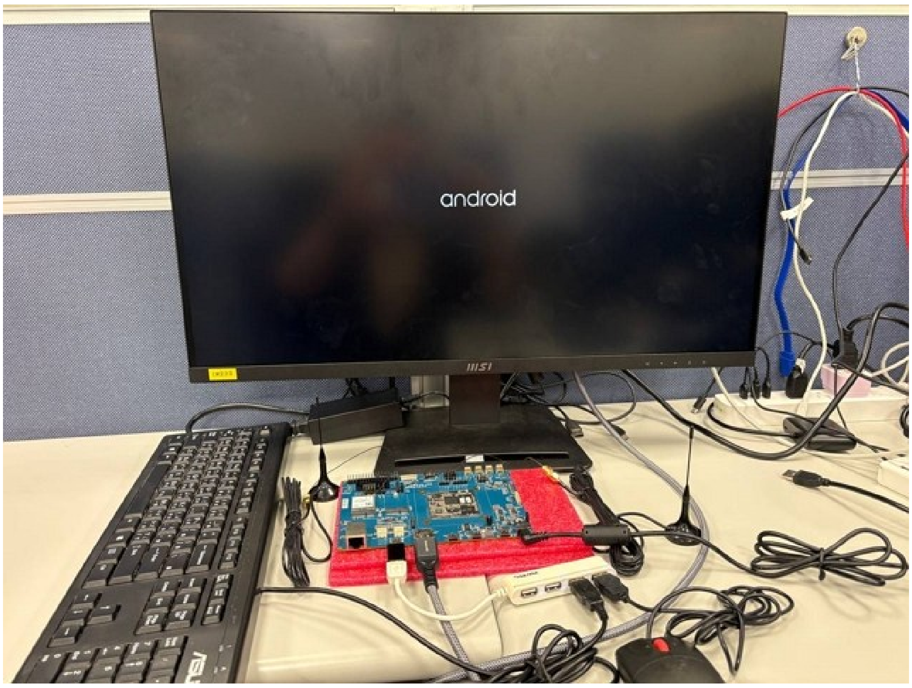
EVB StarterKit Basic Testing

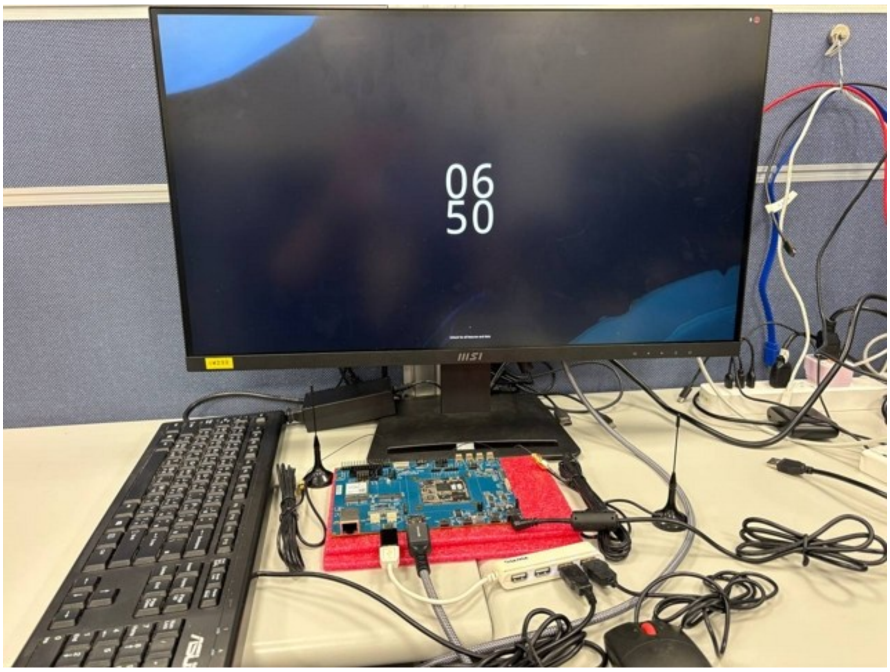
Objective

To verify the flashed image boots up successfully and board's ability to connect to WiFi, access a web browser.

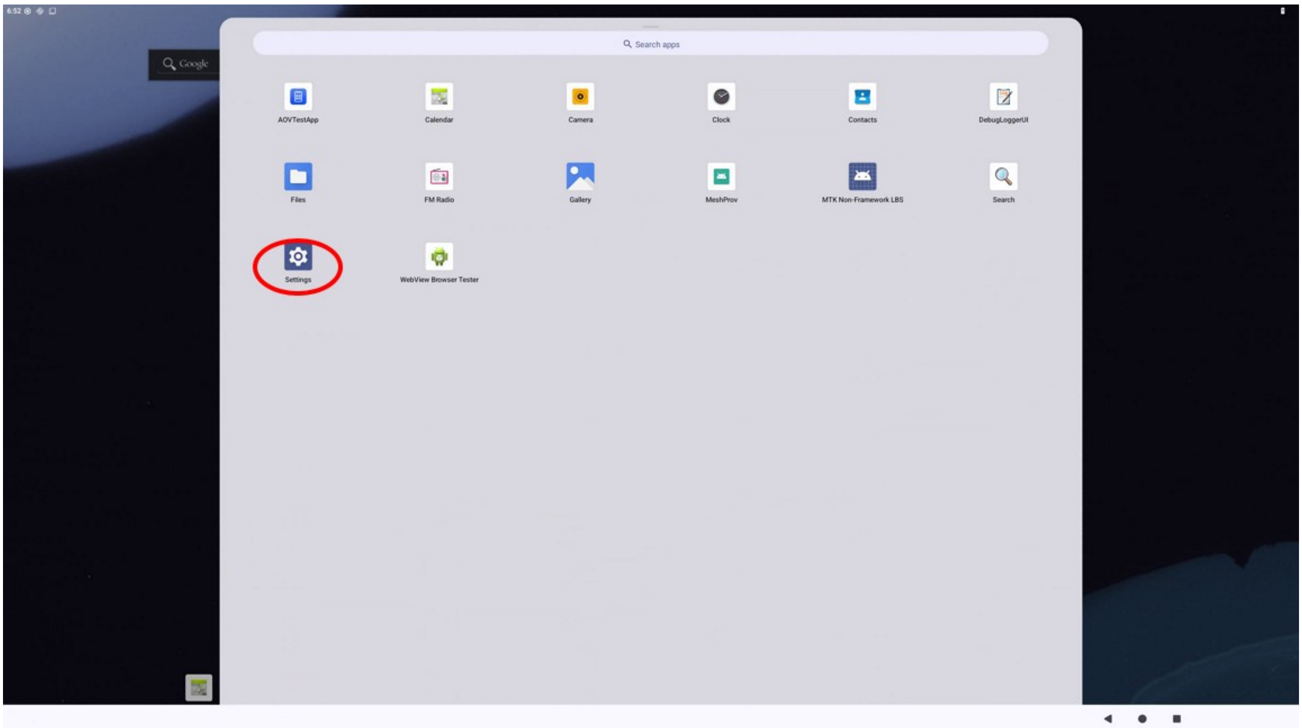
Steps:

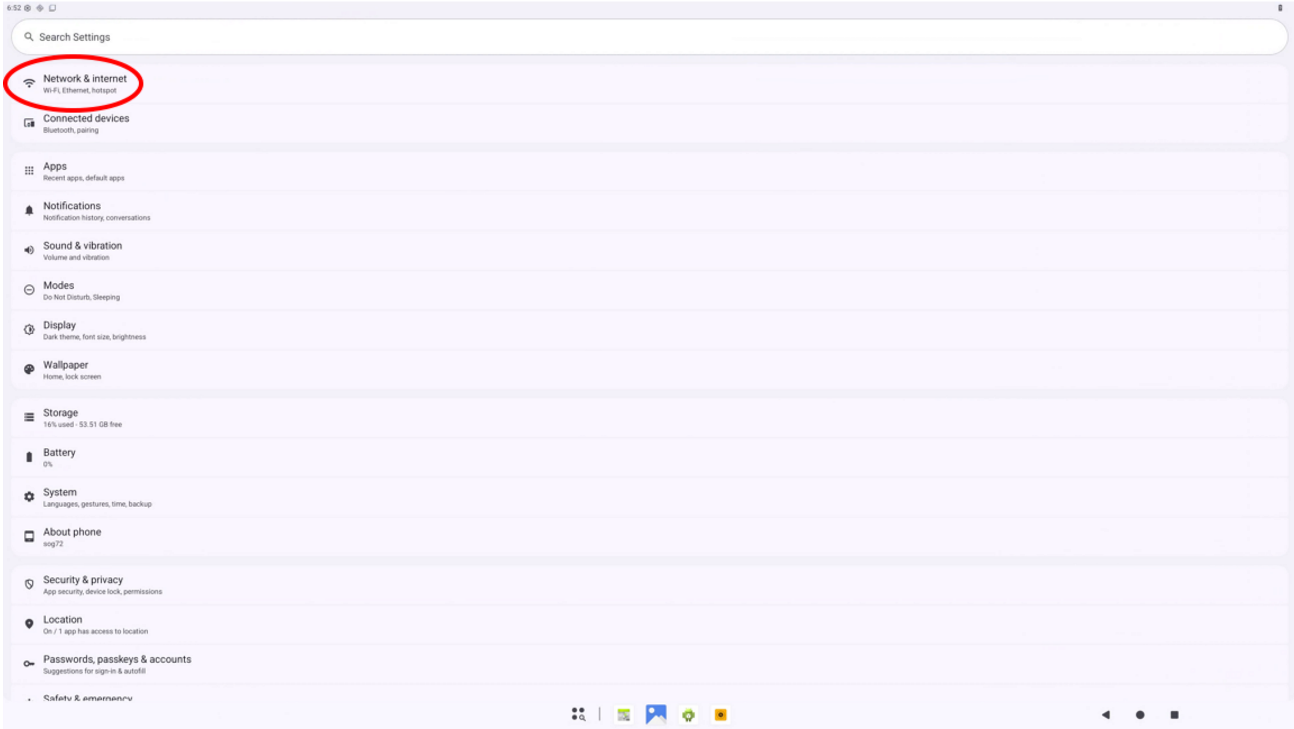
1. **Boot the Board:** Ensure the board is fully powered on and has completed the booting process.



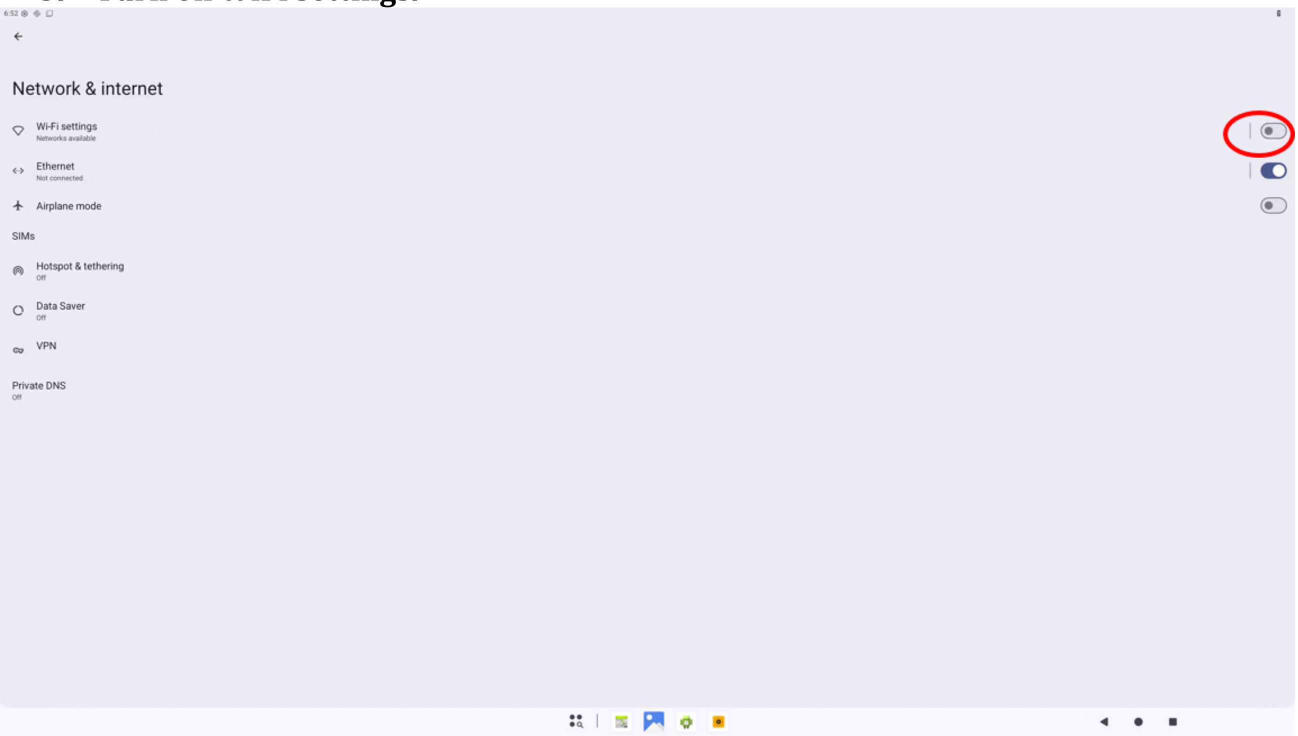


2. **Connect to WiFi:** Navigate to the 'Settings' menu on the board. - Locate the 'WiFi' option and select it. - Choose the desired WiFi network from the list of available networks. - Enter the necessary credentials (e.g., password) to connect to the WiFi network. - Confirm that the board is successfully connected to the WiFi network.

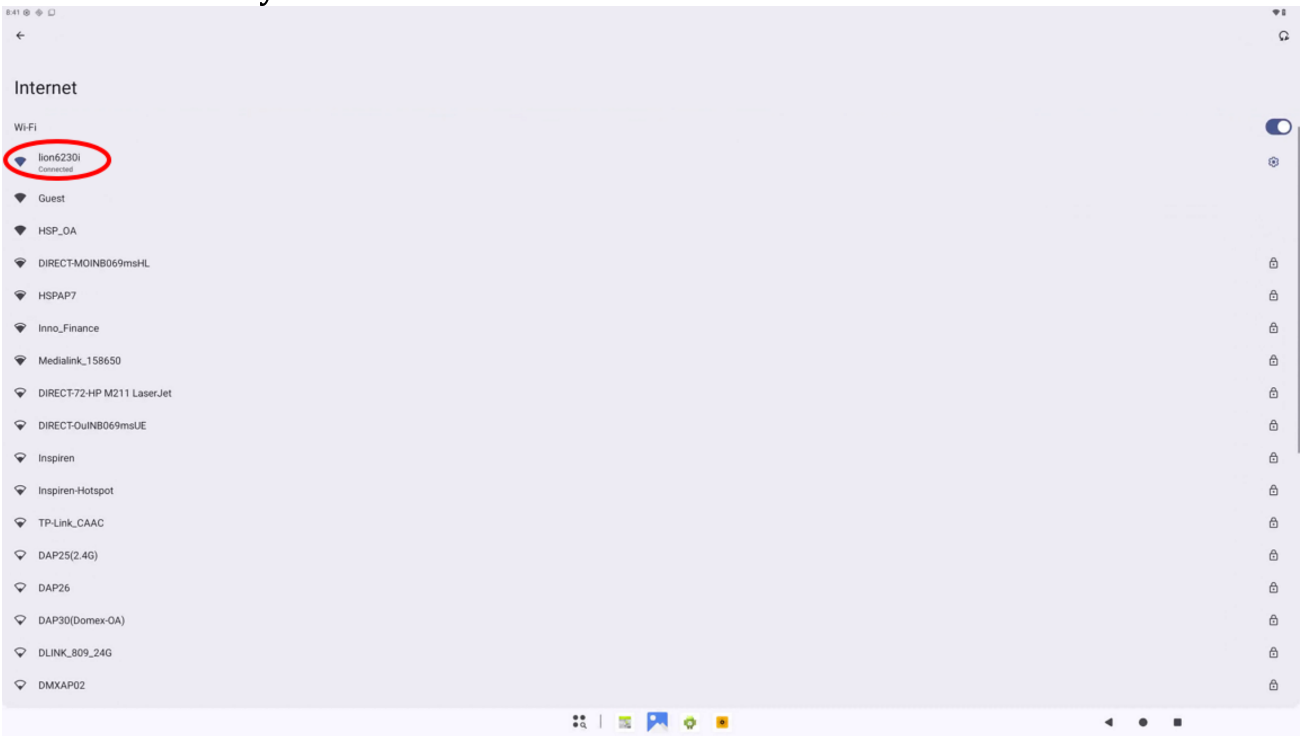




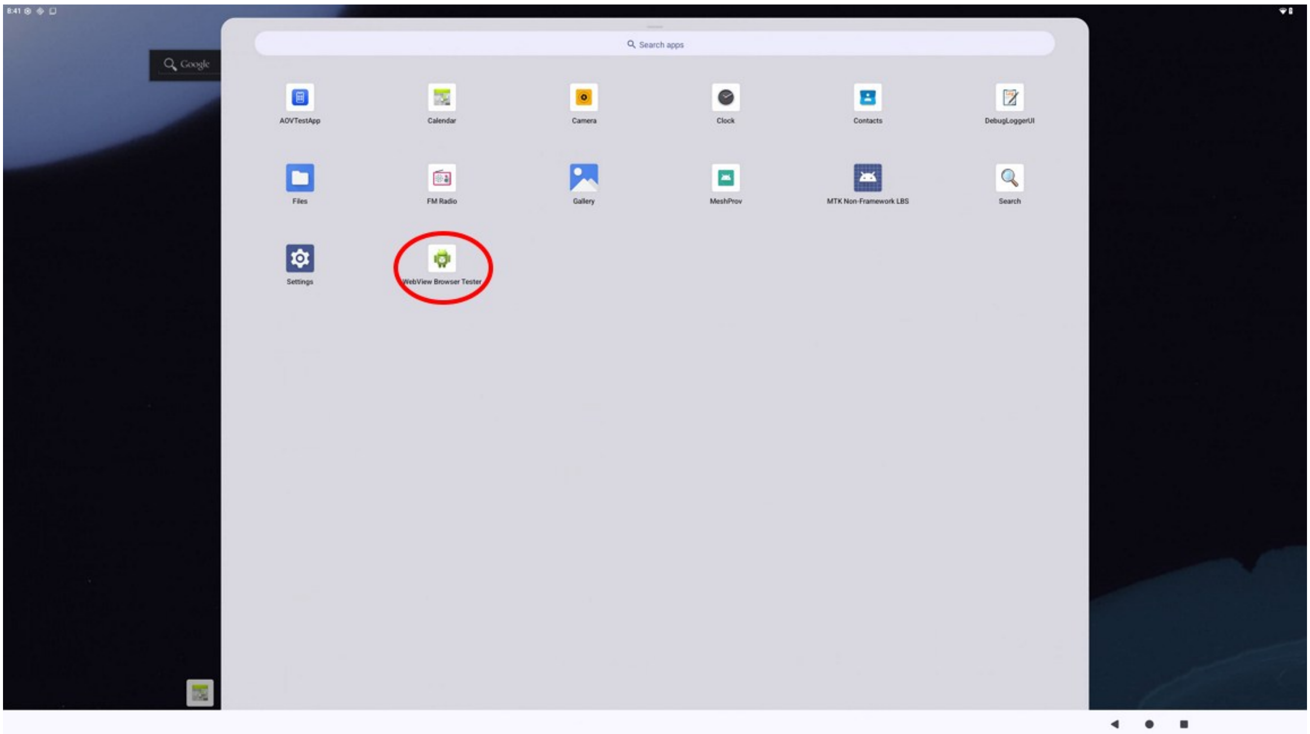
3. Turn on WiFi settings.

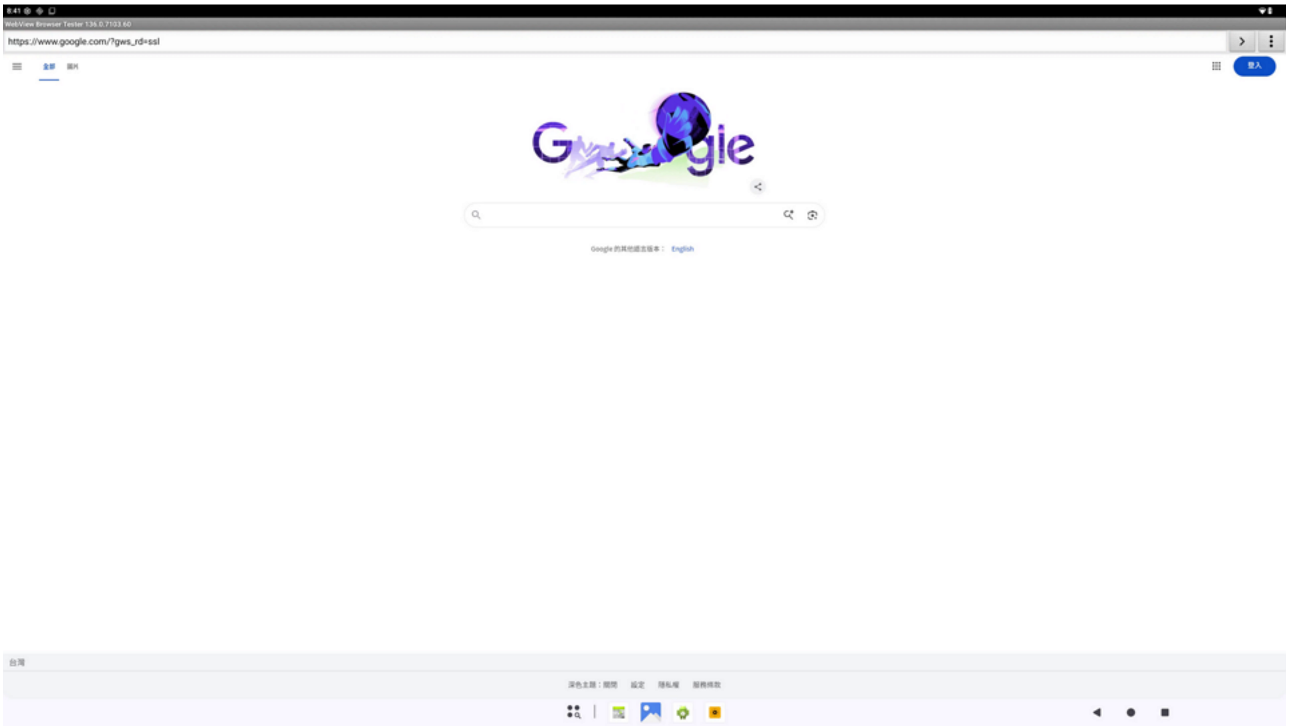


4. Connect to your Wireless AP.



5. **Open the Browser:** Launch the web browser application on the board. - In the browser's address bar, type the URL: www.google.com. - Press 'Enter' or select the 'Go' button to navigate to the Google website.





Note

Ensure that the WiFi network is operational and accessible.